(a)

Tooth detector

one class ‘tooth’; boxes kept if score ≥ 0.3

Numbering model

32 FDI classes; best box per tooth code

Number assignment

each detector box takes the unused numbering box of highest IoU (≥ 0.5); otherwise it stays unassigned

Consistency check

flag for review if a code is duplicated, confidence < 0.5, or the side / arch disagrees with the quadrant; no number is forced

Mask matching

one Mask R-CNN instance per detector box, IoU ≥ 0.5

(b)

